

LAB 5 — Ansible Fundamentals Report

1. Architecture Overview

Ansible Version

```
ansible --version
ansible [core 2.20.2]
  config file = /home/egraba/prog/DevOps-Core-Course/ansible/ansible.cfg
  ansible python module location = /usr/lib/python3.14/site-
packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.14.3
```

Control Node

- OS: Arch Linux
- Ansible installed locally
- SSH key authentication used

Target VM

- Virtualization: KVM/QEMU (virt-manager)
- Network: libvirt NAT (192.168.122.x)
- OS: Ubuntu 25.10 (Questing Quokka)
- Python: `/usr/bin/python3.13`
- Access: SSH (key-based authentication)

```
connection to 192.168.122.221 closed.
egraba@la1n ~/p/D/ansible (lab05)> ssh yan@192.168.122.221
Welcome to Ubuntu 25.10 (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Feb 19 08:37:29 UTC 2026

System load:  0.0                       Processes:    145
Usage of /:   45.4% of 11.21GB          Users logged in: 0
Memory usage: 8%                       IPv4 address for enp1s0: 192.168.122.1
Swap usage:  0%

55 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
```

Infrastructure Overview

```
Host Machine (Arch Linux)
├── Ansible (Control Node)
│   └── SSH
└── KVM / QEMU VM (Ubuntu 25.10)
    ├── Docker Engine
    └── Python Application Container
```

Project Structure

```
ansible/
├── inventory/
│   └── hosts.ini
├── inventory/group_vars/
│   └── all.yml (encrypted via Vault)
├── roles/
│   ├── common/
│   ├── docker/
│   └── app_deploy/
├── playbooks/
│   ├── provision.yml
│   ├── deploy.yml
│   └── site.yml
├── ansible.cfg
├── docs/
│   └── LAB05.md
```

Why Roles Instead of Monolithic Playbooks?

Roles allow:

- Separation of concerns
- Clear modular structure
- Reusability across projects
- Cleaner main playbooks
- Easier debugging and maintenance

Instead of putting all tasks in one large file, each responsibility is isolated.

2. Roles Documentation

2.1 Role: **common**

Purpose

Prepare base Ubuntu system with required tools for further provisioning and automation.

Tasks

- Update apt cache
- Install essential packages

Variables (**defaults/main.yml**)

```
common_packages:
  - python3-pip
  - curl
  - git
  - vim
  - htop
```

Idempotency

- Uses **apt** with **state: present**
- Uses **cache_valid_time** to prevent unnecessary updates
- No imperative shell commands

Handlers

None required.

2.2 Role: **docker**

Purpose

Install and configure Docker Engine.

Tasks

- Install required dependencies
- Add Docker GPG key
- Add Docker repository
- Install Docker packages
- Enable and start Docker service
- Add SSH user to docker group
- Install **python3-docker**

Variables (**defaults/main.yml**)

```
docker_user: yan
```

Handler

```
- name: restart docker
  service:
    name: docker
    state: restarted
```

Triggered when Docker packages change.

Idempotency

- `apt_repository` ensures repo is added once
- `apt state=present`
- `service state=started`
- `user append=yes`

Second run produces zero changes.

2.3 Role: `app_deploy`

Purpose

Deploy containerized Python application from Docker Hub.

Tasks

- Login to Docker Hub (via Vault credentials)
- Pull latest image
- Run container
- Wait for port 5000
- Perform HTTP health check

Variables (Vault — encrypted)

Stored in:

```
inventory/group_vars/all.yml
```

Decrypted content example:

```
dockerhub_username: egrapa
dockerhub_password: <token>
```

```
app_name: devops-core-course-lab2
docker_image: "{{ dockerhub_username }}/{{ app_name }}"
docker_image_tag: latest
app_port: 5000
app_container_name: "{{ app_name }}"
```

Security

- Credentials encrypted with Ansible Vault
- `no_log: true` on login task
- `.vault_pass` excluded from Git
- Only encrypted vault file committed

Idempotency

- `docker_image source: pull`
- `docker_container state: started`
- Container recreated only if configuration changes

3. Idempotency Demonstration

First Run

```
ansible-playbook playbooks/provision.yml
```

Observed:

- Docker repository added
- Packages installed
- Service started

Tasks show `changed`.

```

18 apt_repository:
19   repo: "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu {{ ans...
    ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.
changed: [hyperion]

TASK [docker : Install Docker] *****
changed: [hyperion]

TASK [docker : Ensure Docker running] *****
ok: [hyperion]

TASK [docker : Add user to docker group] *****
changed: [hyperion]

RUNNING HANDLER [docker : restart docker] *****
changed: [hyperion]

PLAY RECAP *****
hyperion                : ok=10  changed=5  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

Second Run

```
ansible-playbook playbooks/provision.yml
```

Observed:

- All tasks show **ok**
- No **changed**

This confirms idempotency.

```

Origin: /home/egraba/prog/DevOps-Core-Course/ansible/roles/docker/tasks/main.yml:19:11

17 - name: Add Docker repository
18   apt_repository:
19     repo: "deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu {{ ans...
    ^ column 11

Use `ansible_facts["fact_name"]` (no `ansible_` prefix) instead.
ok: [hyperion]

TASK [docker : Install Docker] *****
ok: [hyperion]

TASK [docker : Ensure Docker running] *****
ok: [hyperion]

TASK [docker : Add user to docker group] *****
ok: [hyperion]

PLAY RECAP *****
hyperion                : ok=9   changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

Why It Is Idempotent

- Declarative state (**state: present**, **state: started**)
- No shell commands used for state changes
- Ansible checks system state before applying changes

4. Ansible Vault Usage

Vault file:

```
inventory/group_vars/all.yml
```

Encrypted format:

```
$ANSIBLE_VAULT;1.1;AES256  
6238656365...
```

Vault Strategy

- Vault password provided via `--ask-vault-pass`
- `.vault_pass` not committed
- Encrypted file safely committed
- No plaintext secrets in repository

Why Vault Is Important

- Prevents credential leakage
- Secure automation
- Enables safe version control

5. Deployment Verification

Deployment

```
ansible-playbook playbooks/deploy.yml --ask-vault-pass
```

Application deployed successfully.

```
TASK [Gathering Facts] *****
[WARNING]: Host 'hyperion' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of a
discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more informati
ok: [hyperion]

TASK [app_deploy : Login to Docker Hub] *****
changed: [hyperion]

TASK [app_deploy : Pull image] *****
changed: [hyperion]

TASK [app_deploy : Run container] *****
changed: [hyperion]

TASK [app_deploy : Wait for app] *****
ok: [hyperion]

TASK [app_deploy : Health check] *****
ok: [hyperion]

PLAY RECAP *****
hyperion                : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

egrapa@la1n ~/p/D/ansible (lab05)> █
```

Container Status

```
ansible webserver -a "docker ps"
```

Output shows:

```
egrapa/devops-app:latest    Up    0.0.0.0:5000->5000/tcp
```

```
hyperion                : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

egrapa@la1n ~/p/D/ansible (lab05)> ansible webserver -a "docker ps"
[WARNING]: Host 'hyperion' is using the discovered Python interpreter at '/usr/bin/python3.13', but future installation of another Python interpreter could cause a dif
iscovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
hyperion | CHANGED | rc=0 >>
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
af10d70d1759   egrapa/devops-core-course-lab2:late "python app.py"         35 seconds ago Up 34 seconds  0.0.0.0:5000->5000/tcp   devops-core-course-lab2

egrapa@la1n ~/p/D/ansible (lab05)> █
```

Health Check

```
curl http://192.168.122.221:5000/health
```

Response:

```
af10d70d1759   egrapa/devops-core-course-lab2:late "python app.py"         35 seconds ago Up 34
egrapa@la1n ~/p/D/ansible (lab05)> curl http://192.168.122.221:5000/health
{"status": "healthy", "timestamp": "2026-02-26T10:42:51.977415+00:00", "uptime_seconds": 50}
egrapa@la1n ~/p/D/ansible (lab05)> █
```

Application is accessible and running.

6. Key Decisions

Why Use Roles?

To separate infrastructure concerns and maintain clean modular automation.

How Do Roles Improve Reusability?

Each role can be reused independently across different projects.

What Makes a Task Idempotent?

It declares a desired state and does nothing if the system already matches that state.

How Do Handlers Improve Efficiency?

They run only when notified, preventing unnecessary service restarts.

Why Is Ansible Vault Necessary?

To protect Docker Hub credentials from exposure in Git repositories.

7. Challenges Encountered

- Vault variables not being loaded due to incorrect group_vars location
- Ensuring correct ansible.cfg was used
- Docker Hub authentication token configuration
- Understanding Ansible variable precedence
- Ensuring idempotency of repository and Docker installation tasks